

Temporal Environmental Variability Drives the Evolution of Cooperative Breeding in Birds

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Table S1. Social Systems and Environmental Characters of African Starlings

Genus	Species	Social System	Habitat	P	C	M
<i>Cinnyricinclus</i>	<i>leucogaster</i>	Cooperative	Savanna	0.47	0.23	0.24
<i>Grafisia</i>	<i>torquata</i>	Noncooperative	Savanna	0.56	0.28	0.28
<i>Hartlaubius</i>	<i>aurata</i>	Noncooperative	Forest	-	-	-
<i>Hylopsar</i>	<i>cupreocauda</i>	Noncooperative	Forest	0.51	0.29	0.22
<i>Hylopsar</i>	<i>purpureiceps</i>	Noncooperative	Forest	0.49	0.31	0.18
<i>Lamprotornis</i>	<i>acuticaudus</i>	Noncooperative	Savanna	0.58	0.26	0.31
<i>Lamprotornis</i>	<i>albicapillus</i>	Cooperative ^a	Savanna	0.43	0.27	0.16
<i>Lamprotornis</i>	<i>australis</i>	Noncooperative	Savanna	0.46	0.23	0.22
<i>Lamprotornis</i>	<i>bicolor</i>	Cooperative	Savanna	0.35	0.22	0.12
<i>Lamprotornis</i>	<i>caudatus</i>	Cooperative	Savanna	0.63	0.31	0.32
<i>Lamprotornis</i>	<i>chalcurus</i>	Noncooperative	Savanna	0.57	0.31	0.26
<i>Lamprotornis</i>	<i>chalybaeus</i>	Noncooperative	Savanna	0.47	0.23	0.23
<i>Lamprotornis</i>	<i>chloropterus</i>	Cooperative	Savanna	0.50	0.23	0.26
<i>Lamprotornis</i>	<i>fischeri</i>	Cooperative	Savanna	0.41	0.21	0.20
<i>Lamprotornis</i>	<i>hildebrandti</i>	Cooperative ^a	Savanna	0.41	0.19	0.21
<i>Lamprotornis</i>	<i>iris</i>	Cooperative	Savanna ^b	0.58	0.20	0.38
<i>Lamprotornis</i>	<i>mevesii</i>	Cooperative	Savanna	0.51	0.23	0.27
<i>Lamprotornis</i>	<i>nitens</i>	Cooperative	Savanna	0.42	0.21	0.21
<i>Lamprotornis</i>	<i>ornatus</i>	Noncooperative	Forest	0.52	0.31	0.21
<i>Lamprotornis</i>	<i>pulcher</i>	Cooperative	Savanna	0.57	0.31	0.26
<i>Lamprotornis</i>	<i>purpureus</i>	Noncooperative	Savanna	0.57	0.29	0.27
<i>Lamprotornis</i>	<i>purpureoptera</i>	Cooperative	Savanna	0.52	0.34	0.17
<i>Lamprotornis</i>	<i>regius</i>	Cooperative	Savanna	0.43	0.24	0.19
<i>Lamprotornis</i>	<i>shelleyi</i>	Noncooperative	Savanna	0.46	0.22	0.24
<i>Lamprotornis</i>	<i>splendidus</i>	Noncooperative	Forest	0.53	0.29	0.23
<i>Lamprotornis</i>	<i>superbus</i>	Cooperative	Savanna	0.44	0.24	0.20
<i>Lamprotornis</i>	<i>unicolor</i>	Cooperative	Savanna	0.52	0.24	0.28
<i>Neocichla</i>	<i>gutturialis</i>	Cooperative	Savanna	0.55	0.21	0.33
<i>Notopholia</i>	<i>corruscus</i>	Noncooperative	Forest ^c	0.36	0.20	0.16
<i>Onychognathus</i>	<i>albirostris</i>	Noncooperative	Desert ^d	0.41	0.17	0.23
<i>Onychognathus</i>	<i>blythii</i>	Noncooperative	Desert ^d	0.49	0.32	0.17
<i>Onychognathus</i>	<i>frater</i>	Noncooperative	Desert ^d	-	-	-
<i>Onychognathus</i>	<i>fulgidus</i>	Noncooperative	Forest	0.49	0.29	0.20
<i>Onychognathus</i>	<i>morio</i>	Noncooperative	Savanna	0.41	0.21	0.20
<i>Onychognathus</i>	<i>nabourop</i>	Noncooperative	Desert	0.41	0.25	0.15
<i>Onychognathus</i>	<i>salvadorii</i>	Noncooperative	Desert ^d	0.37	0.22	0.15
<i>Onychognathus</i>	<i>neumanni</i>	Noncooperative	Forest	0.61	0.29	0.32
<i>Onychognathus</i>	<i>tenuirostris</i>	Noncooperative	Forest ^e	0.45	0.23	0.21
<i>Onychognathus</i>	<i>walleri</i>	Noncooperative	Forest	0.47	0.26	0.21
<i>Poeoptera</i>	<i>femoralis</i>	Noncooperative	Forest	0.45	0.24	0.20
<i>Poeoptera</i>	<i>kenricki</i>	Noncooperative	Forest	0.38	0.17	0.20
<i>Poeoptera</i>	<i>lugubris</i>	Noncooperative	Forest	0.54	0.30	0.23
<i>Poeoptera</i>	<i>sharpii</i>	Noncooperative	Forest	0.47	0.29	0.17
<i>Poeoptera</i>	<i>stuhlmanni</i>	Noncooperative	Forest	0.50	0.34	0.16
<i>Speculipastor</i>	<i>bicolor</i>	Cooperative ^a	Savanna	0.46	0.26	0.19

Social systems were divided into cooperative and noncooperative. Habitats were divided into savanna and nonsavanna (i.e., desert and forest). Two species known to show facultative cooperative breeding (*C. leucogaster* and *L. hildebrandti*) were classified as cooperative in our analyses. Mean values of Colwell's measures of predictability (P), constancy (C), and contingency (M) for each species are also listed.

^a D.R.R. personal observation.

^b This species might also inhabit forest-savanna mosaics, but we classify its primary habitat as savanna.

^c This species inhabits thickets and riverine and coastal forests, which we classify as forest.

^d These species are typically found in arid, rocky highland habitats, which we classify as desert.

^e Because this species is associated with riverine gorges, which are usually, but not always, in forests, we classify its primary habitat as forest.